Steering Through the Trees: Physical Lie Vector Field Evasion of Tree Ensembles in Network Intrusion Detection

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Abstract

Gradient-boosted decision trees dominate flow-based network intrusion detection, yet their robustness against physically realizable problem-space evasion remains an open question. Feature-space attacks ignore coupling invariants between flow features and produce unrealizable telemetry, whereas stochastic packet mutators incur heavy query and perturbation overhead. We propose Flow-Field Evasion (FFE), which formalizes four protocol-compliant actions (temporal dilation, payload padding, chaff injection, and TCP window morphing) as infinitesimal generators of a Lie cone on the physical flow manifold. FFE steers along the cost-effective Lie generator and snaps to tree split thresholds via boundary snapping (TSBS). Evaluated across 12,000-flow real benchmarks from CSE-CIC-IDS2018, UNSW-NB15, and CIC-IDS2017 against CUDA XGBoost and LightGBM, FFE achieves 59.0%–97.0% Real ASR on CSE-CIC-IDS2018 and up to 98.0% on UNSW-NB15 (98.0%–100.0% Physical Feasibility Rate), matching Particle Swarm Optimization in suc-

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cess (59.0% vs. 54.0%) at 7,500× lower perturbation cost ($\bar{D}_{L_1} = 0.392$ vs. 2,946.18) and half the query budget. Conversely, on CIC-IDS2017 deep immutable feature partitions restrict evasion to 3.0%–11.0% (93.0% PFR). Furthermore, standalone malleability penalties inadvertently expose unpenalized features (surging evasion from 51.6% to 98.9%), whereas dual hardening (penalties plus Lie-guided adversarial training) slashes evasion to 5.3% with zero clean accuracy degradation.

Keywords: Adversarial machine learning, Network intrusion detection, Gradient boosted decision trees, Evasion attacks, Problem-space constraints, Lie algebra, Adversarial training

1. Introduction

Modern network intrusion detection systems (NIDS) increasingly rely on machine learning over flow-based telemetry. At multi-gigabit rates, inspecting every packet payload is prohibitive, so sensors such as CICFlowMeter [1] aggregate bidirectional packet exchanges into tabular flow records of timing, length, header and flag statistics. Gradient-boosted decision trees (GBDTs), notably XGBoost [2], LightGBM [3] and CatBoost [4], are the dominant classifiers for these records.

Near-perfect accuracy on benchmark corpora does not guarantee robustness in operation. Closed-world evaluations reward models that exploit testbed artifacts [5] and shortcut features [6]. Attackers, however, control the malleable parts of their traffic, such as timing, padding and window sizes, so a detector that relies on these features can be evaded without losing the exploit [7].

Evading a tree ensemble is hard for two reasons. First, the ensemble is piecewise constant, so $\nabla_x F(x) = 0$ for almost all x and gradient-based attacks receive no signal. Second, flow features obey conservation laws. Flow Bytes/s satisfies $x_8 = 10^6(x_4 + x_5)/x_6$, where x_4 and x_5 are the forward and backward byte totals and x_6 is the duration in microseconds. A perturbation that changes the byte totals without updating the rate produces a phantom flow that protocol sanity checks reject [8, 7].

Existing work follows one of two paradigms, and neither is adequate. Feature-space methods such as ZOO [9] and HopSkipJump [10] ignore the physical coupling: when evaluated against network protocol invariants, their Physical Feasibility Rate (PFR) collapses to 0.0%–6.0%, yielding 0.0% Real ASR despite reporting up to 100% mathematical evasion in unconstrained coordinates. Heuristic multi-objective search (CAA/MOEVA) [12, 13] achieves only 1.0% Real ASR (15.0% PFR) after 336.2 queries per flow. Conversely, packet-level trial-and-error tools such as Traffic Manipulator PSO [11] preserve physical feasibility (100.0% PFR), but query the classifier blindly as a black box, requiring 182.6 queries per flow and achieving 54.0% Real ASR at severe perturbation overhead ($\bar{D}_{L_1} = 2,946.18$).

This article proposes Flow-Field Evasion (FFE), which combines both properties. FFE derives physical perturbations as differential generators, so the continuous flow satisfies the invariants exactly and a projection enforces them on the discrete schema, and it steers flows using the split geometry of the ensemble. To our knowledge, no prior attack on tree ensembles combines these two properties.

The main contributions are as follows.

- We model four packet-level attacker actions (temporal dilation, payload padding, chaff injection and TCP window morphing) as closed-form infinitesimal generators of a Lie cone on the flow manifold.

- We prove that trajectories in this cone preserve the rate, mean-length, header and MTU invariants of flow records in the continuous relaxation of the flow manifold.

- We present FFE, which moves each flow along the generator with the largest cost-weighted score decrease and finishes with tree-split boundary snapping (TSBS).

- We evaluate FFE on real benchmark traffic from CSE-CIC-IDS2018, UNSW-NB15, and CIC-IDS2017. Against CUDA XGBoost on CSE-CIC-IDS2018, FFE achieves 59.0% Real ASR with 100% PFR, matching Traffic Manipulator PSO in evasion efficacy (59.0% vs. 54.0%) while requiring less than half the query complexity (79.3 vs. 182.6) and $7,500\times$ lower perturbation overhead ($\bar{D}_{L_1} = 0.392$ vs. 2,946.18). Across architectures and datasets, FFE reaches up to 94.0%–98.0% Real ASR on UNSW-NB15 and LightGBM, but is constrained to 3.0%–11.0% on CIC-IDS2017 due to deep immutable feature partitions. Furthermore, we reveal that standalone feature malleability penalization creates a vulnerability trap (surging evasion to 98.9%), whereas dual hardening combining malleability penalties with Lie-guided adversarial training (MPS + LGAT) slashes FFE evasion to 5.3% (94.7% robust recall) with zero clean accuracy degradation.

2. Related Work and Problem Formulation

2.1. Related Work and State-of-the-Art Taxonomy

Gradient-boosted decision trees (GBDTs), including XGBoost [2], LightGBM [3] and CatBoost [4], are the standard choice for flow-based network intrusion detection on benchmarks such as CSE-CIC-IDS2018 [1], UNSW-NB15 [14] and CIDDS-001 [15]. Comprehensive empirical studies support this choice [16, 17, 7]. Their split rules depend only on the rank order of feature values and are cheap to evaluate, which makes them attractive for deployment. Learned detectors have long been questioned for operation outside the closed world of their training data [5], but robustness to realistic evasion has received less attention.

Early adversarial attacks on NIDS were transplanted from image classification. Szegedy et al. [18], Goodfellow et al. [19] and Papernot et al. [20] showed that gradient-based perturbations fool deep networks, and optimizers such as the Carlini-Wagner L_2 attack [21] were then applied to tabular flow features. These studies reported high success rates, but the perturbed vectors were not network flows. Continuous optimization in Euclidean space assigns fractional values to integer packet counts, modifies immutable fields such as destination port, produces negative durations, and breaks the conservation between byte totals and packet counts. A feature vector that no valid packet sequence can produce says nothing about the deployed detector.

The problem-space view corrects this. Pierazzi et al. [8] showed that an adversarial perturbation is meaningful only when it is realized as a transformation of the physical object, such as a binary or a packet sequence, and that the transformation must preserve functionality and survive the fea-

ture extractor. Verkerken et al. [22] apply this view to defining host-space adversarial perturbations against NIDS. Surveys and audits of this literature [7, 23, 24, 25] argue that many feature-space attacks assume threat models that cannot be realized with valid packets under RFC protocol specifications, so their success rates overstate the risk once evaluated against true packet parsers and middleboxes. Constrained tabular attacks address feasibility only in part. CAA [12] combines a gradient-based attack (CAPGD) with MOEVA [13], a multi-objective genetic search whose constraint penalties are heuristic and whose population search receives no directional guidance. TabularBench [26] standardizes such evaluations across deep and tree models. Packet-level tools avoid these problems by operating on real traffic. Traffic Manipulator [11] uses particle swarm optimization to mutate PCAP files by perturbing inter-packet intervals, adding payload bytes and injecting chaff packets. IDSGAN [27] generates adversarial flow records with generative adversarial networks, but its feasibility is only approximated in feature space.

Exact methods exploit the piecewise-constant structure of trees. Kantchelian et al. [28] cast minimal evasion against tree ensembles as a mixed-integer linear program. The formulation is exact but NP-hard in general and accepts only linear constraints, so it cannot express the nonlinear rate relations between bytes and duration that flow records obey. SAT and MaxSAT encodings of tree ensembles [29, 30] provide formal explanations and robustness checks, and LT-Attack [31] attacks tree ensembles through continuous leaf approximations. The MILP and LT-Attack operate in unconstrained feature space and do not enforce coupled network invariants. Robust training

of tree ensembles [32, 33, 34] and adversarial retraining of neural NIDS [35] improve robustness within the same feature-space threat model. Black-box query methods such as ZOO [9] and HopSkipJump [10] estimate the decision boundary by sampling the classifier's output, and population or swarm searches mutate parameters without knowing which splits are active.

Table 1 summarizes these paradigms and the gap they leave. Methods that exploit the structure of tree ensembles or work in unconstrained feature space do not produce flows that survive packet parsing, while methods that preserve physical feasibility treat the detector as a black box and need hundreds to thousands of queries per flow. Flow-Field Evasion (FFE) addresses this gap by steering flows along infinitesimal packet-level generators that keep every step on the flow manifold, and by using the split geometry of the trees to choose the direction.

2.2. Valid Packet Streams and Flow Telemetry

A packet stream $P = ((p_1, t_1), \ldots, (p_m, t_m))$ is an ordered sequence of packets with $0 \leq t_1 \leq \cdots \leq t_m$, where t_i is the transmission time at the network interface card. Only streams in the admissible set $\mathcal{P}_{\text{valid}}$ are considered. Every packet must carry valid IPv4, TCP and UDP checksums [36, 37, 38] and obey the interface MTU:

$$\forall i \in \{1, \ldots, m\}, \quad \text{Len}_{\text{IP}}(p_i) \leq \text{MTU}_{\text{IP}} = 1500 \text{ bytes} \tag{1}$$

A flow is the bidirectional set of packets sharing a 5-tuple, split into forward (client-to-server) and backward packets. A flow exporter such as CICFlowMeter [1] computes a deterministic, many-to-one mapping

$$\Phi : \mathcal{P}_{\text{valid}} \to \mathbb{R}^d \tag{2}$$

Table 1: Representative evasion, verification and defense paradigms for tabular and network telemetry. Physical feasibility reports the empirical PFR of this benchmark where it was evaluated (Section 4.2).

Method	**Target model**	**Engine**	**Queries**	**Coupled invariants**	**Physical feasibility**
Kantchelian et al. [28]	GBDT, random forest	MILP (worst case $2^{\mathcal{O}(DT)}$)	0 (white-box)	Unsupported (linear only)	Unconstrained
Zhang et al. [31] (LT-Attack)	Tree ensembles	Continuous leaf approximation	$O(N_{\text{iter}})$	Not modeled	Unconstrained
Ignatiev et al. [29]; Izza et al. [30]	GBDT, decision trees	MaxSAT / SAT verification	0 (white-box)	Not modeled	Not applicable (explanation)
Chen et al. (ZOO) [9]	Black-box classifiers	Zeroth-order coordinate descent	$O(d \cdot N_{\text{iter}})$	Not modeled	6.0% (empirical)
Chen et al. (HSJ) [10]	Black-box decision	Boundary random walk	$O(10^3)$	Not modeled	0.0% (empirical)
Lin et al. (IDS-GAN) [27]	Black-box NIDS	Generative adversarial network	$O(N_{\text{gen}})$	Learned distribution	Approximated (feature space)
Ghamizi et al. (MOEVA) [13]	Constrained tabular	Multi-objective genetic search	$\leq$1,250	Heuristic penalty	Partial (15.0% empirical)
Simonetto et al. (CAA) [12]	Tabular models	CAPGD with MOEVA	1,240	Approximate penalty	Partial (15.0% empirical)
Han et al. (Traffic Manipulator) [11]	Black-box NIDS	PSO on PCAP files	$O(10^3$–$10^4)$	Packet re-evaluation	100% (packet mutation)
Flow-Field Evasion (ours)	**XGBoost, LightGBM**	**Lie vector field steering**	$\leq 4M_{\max}+1$ **probes plus TSBS (mean 79.3)**	**Exact closed-form**	**100.0% (schema-checked)**

Note: T trees, D depth, $M_{\max} = 20$, $K = 4$ generators. Empirical PFR values are from Table 2.

and $x = \Phi(P)$ summarizes the transmission.

2.3. Twelve-Dimensional Feature Schema

We use a standardized 12-dimensional schema of the core metrics that state-of-the-art models analyze on CSE-CIC-IDS2018 [1]. The feature vector $x \in \mathbb{R}^{12}$ comprises: x_0, destination port; x_1, protocol (6 for TCP, 17 for UDP); x_2, number of forward packets; x_3, number of backward packets; $x_4 = \sum_{p_i \in P_{\text{fwd}}} \text{Len}(p_i)$, forward wire bytes including IP and transport headers; x_5, backward bytes; $x_6 = (t_m - t_1) \times 10^6$, flow duration in microseconds; x_7, initial forward TCP window ($0 \le x_7 \le 65535$, and $x_7 \equiv 0$ for UDP); x_8, flow bytes per second; x_9, flow packets per second; x_{10}, mean forward packet length; and x_{11}, forward transport header bytes, which must be at least the per-packet minimum header size times x_2. The rates and mean are fixed by the relation

$$x_8 = g_8(x_4, x_5, x_6) = \frac{10^6\,(x_4 + x_5)}{x_6}, \qquad x_9 = g_9(x_2, x_3, x_6) = \frac{10^6\,(x_2 + x_3)}{x_6},$$
$$x_{10} = g_{10}(x_4, x_2) = \frac{x_4}{\max(1, x_2)} \tag{3}$$

2.4. Feature Invariants

The indices $\mathcal{D} = \{0, \ldots, 11\}$ fall into five constraint classes derived from network semantics, and a feature may belong to more than one class.

- $\mathcal{I}_{\text{imm}} = \{0, 1, 3, 5\}$: port, protocol and backward packets and bytes are fixed by the service and the server, so $\Delta x_i = 0$.

- $\mathcal{I}^{+}_{\text{mono}} = \{2, 4, 6, 11\}$: an attacker may add packets, pad payloads or delay transmissions, but cannot remove bytes the payload needs, so

$\Delta x_j \geq 0$.

- $\mathcal{I}_{\text{bound}} = \{7\}$: the window stays within $0 \leq x_7 + \Delta x_7 \leq 65535 \cdot \mathbb{I}_{x_1=6}$.
- $\mathcal{I}_{\text{dep}} = \{8, 9, 10\}$: these features are the deterministic functions g_8, g_9, g_{10} of the base metrics given above.
- $\mathcal{I}_{\text{ineq}} = \{4, 11\}$: $x_{11} \geq x_2(20\,\mathbb{I}_{x_1=6} + 8\,\mathbb{I}_{x_1=17})$ and $0 \leq x_4 \leq 1500x_2$.

Feature x_4 is both monotone and MTU bounded. The set $\mathcal{I}^+_{\text{mono}}$ gives the mutable monotone features that Algorithm 2 modifies.

2.5. Flow Manifold and Ambient Relaxation

The physical flow space is the discrete image of valid packet streams under the telemetry exporter:

$$\mathcal{M}_{\text{flow}} = \left\{ x \in \mathbb{R}^{12} \;\middle|\; \exists P \in \mathcal{P}_{\text{valid}} \text{ s.t. } \Phi(P) = x \right\} \subset \mathbb{Z}^9_{\geq 0} \times \mathbb{R}^3_{\geq 0} \tag{4}$$

where discrete coordinates $(x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_{11})$ satisfy integrality (with packet counts $x_2 \in \mathbb{N}_{\geq 1}, x_3 \in \mathbb{N}_{\geq 0}$, duration in integer microseconds $1 \leq x_6 \leq 1.1 \times 10^8\,\mu$s bounded below the 120-second exporter timeout), rates $(x_8, x_9, x_{10}) \in \mathbb{R}^3_{\geq 0}$, and packet lengths comply with interface MTU restrictions ($x_4 \leq 1500x_2$). We focus on IPv4 ($L_{\text{IPv4}} = 20$ bytes) and standard Ethernet MTU.

To enable continuous differential geometric optimization, we define the *continuous ambient flow manifold* $\widetilde{\mathcal{M}}_{\text{flow}} \subset \mathbb{R}^{12}$ by relaxing packet counts and duration to positive reals ($x_2 \geq 1, x_6 > 0$), with transport header limits

$H_{\min}(x_1) = 20 \cdot \mathbb{I}_{x_1=6} + 8 \cdot \mathbb{I}_{x_1=17}$ and $H_{\max}(x_1) = 60 \cdot \mathbb{I}_{x_1=6} + 8 \cdot \mathbb{I}_{x_1=17}$:

$$\widetilde{\mathcal{M}}_{\text{flow}} = \left\{ x \in \mathbb{R}^{12} \;\middle|\; \begin{array}{l} x_0 \in [1, 65535], \quad x_1 \in \{6, 17\}, \quad x_2 \geq 1, \quad x_3, x_5 \geq 0, \\ H_{\min}(x_1) x_2 \leq x_{11} \leq H_{\max}(x_1) x_2, \\ 0 \leq x_4 \leq 1500 x_2, \quad 0 < x_6 \leq 1.1 \times 10^8, \\ 0 \leq x_7 \leq 65535 \cdot \mathbb{I}_{x_1=6}, \quad x_{10} = x_4 / x_2, \\ x_8 = 10^6 (x_4 + x_5)/x_6, \quad x_9 = 10^6 (x_2 + x_3)/x_6 \end{array} \right\} \tag{5}$$

Continuous vector field flows are formulated on $\widetilde{\mathcal{M}}_{\text{flow}}$, while discrete execution snaps continuous trajectories back to $\mathcal{M}_{\text{flow}}$ via the closed-form projection operator $\mathcal{P}_{\mathcal{M}}$ (Section 3.4).

2.6. Threat Model

The adversary starts from a malicious flow $x_{\text{orig}} = \Phi(P) \in \mathcal{M}_{\text{flow}}$ classified as an attack, $P(\text{Malicious} \mid x_{\text{orig}}) \geq 0.5 \iff F(x_{\text{orig}}) \geq 0$, and seeks a modified flow $x^* = \Phi(P^*) \in \mathcal{M}_{\text{flow}}$ with

$$P(\text{Malicious} \mid x^*) < 0.5 \iff F(x^*) < 0 \tag{6}$$

Tier 1 (white-box) grants access to the ensemble architecture, leaf weights and split thresholds Θ_j, which enables exact Tree-Split Boundary Snapping (TSBS) and forms the primary benchmark (Section 4.2). Tier 2 (black-box) allows only query access or zero-knowledge transfer of flows crafted on a surrogate ensemble, which Table 5 evaluates.

The transformed trace P^* must preserve the malicious semantics, $\mathcal{S}(P^*) = \mathcal{S}(P)$, so that exploit delivery, shellcode execution, credential cracking or service degradation still succeeds on the victim. Payload padding preserves

discrete payload exploits by appending non-executable, protocol-compliant bytes (SSH ignore messages, RFC 4253 [39]). Temporal dilation still delivers every packet, although it lowers the burst intensity of volumetric floods (Section 6.1); we do not test that the attack's effect on the victim is preserved.

The NIDS is an ingress biflow sensor (for example, an IPFIX collector at a border router or DMZ firewall). The adversary controls only the client endpoint, so it changes the forward features $x_2, x_4, x_6, x_7, x_{11}$ and the derived rates, while the server-generated x_3 and x_5 are immutable. Forward chaff is sent as pure ACKs with seq = SND.NXT and ack = RCV.NXT. A TCP endpoint never acknowledges a pure ACK (RFC 9293 [37], § 3.8.1), so these packets elicit no backward packets or bytes and $\Delta x_3 = \Delta x_5 = 0$. Chaff is spaced or interleaved with forward payload and window updates so that three duplicate ACKs do not trigger fast retransmit (RFC 5681 [40], § 3.2). Window morphing edits the window field of the initial SYN consistently with window scaling (RFC 7323 [41]). For encrypted protocols such as SSH, padding is inserted at the application layer before cryptographic encapsulation (RFC 4253 [39], § 11.2).

The adversary minimizes the total operational cost

$$\min \quad \mathcal{C}(x^*, x_{\text{orig}}) = \sum_{k=1}^{4} c_k s_k \tag{7}$$

where s_k is the amount of generator k applied and c_k is its unit cost in transmission latency, bandwidth expansion and socket buffer allocation.

3. The Flow-Field Evasion (FFE) Framework

Figure 1 summarizes the framework.

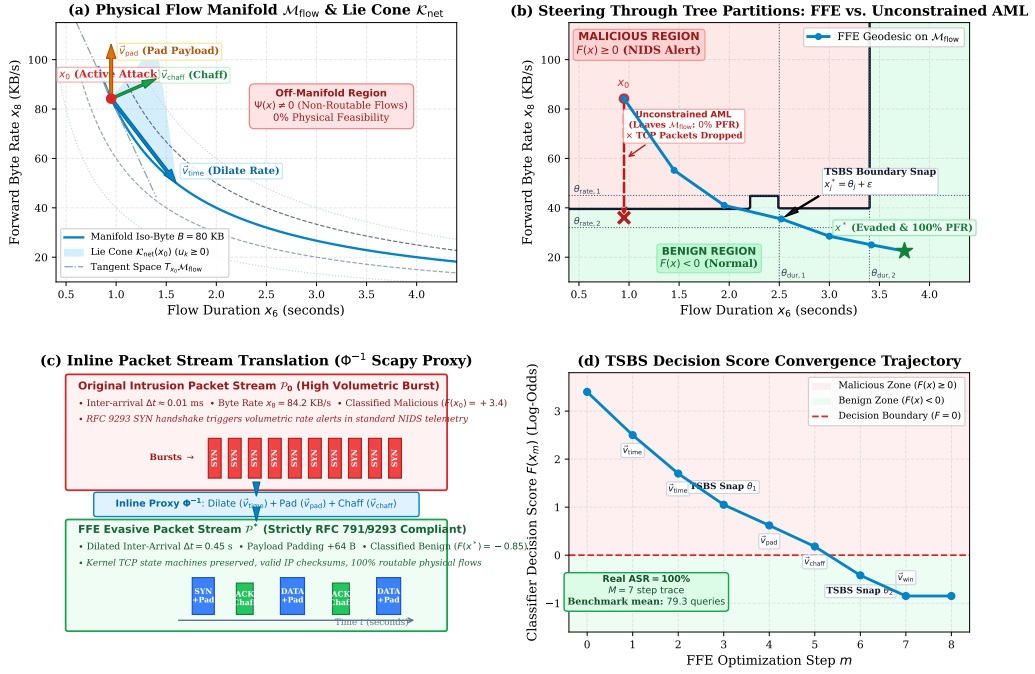


Figure 1: Architecture of the Flow-Field Evasion (FFE) framework. **(a)** The physical conservation constraints define the flow manifold $\mathcal{M}_{\text{flow}}$, and the generators span a forward Lie cone that keeps trajectories on it. **(b)** Axis-aligned tree partitions push unconstrained feature-space perturbations off the manifold, whereas FFE moves along the manifold and finishes with tree-split boundary snapping (TSBS). **(c)** Packet-level realization of the flow transformations through an inline proxy. **(d)** Decision score descent over the steps of an attack trace.

3.1. Geometry of the Flow Manifold and Its Generators

The physical flow set $\mathcal{M}_{\text{flow}} = \Phi(\mathcal{P}_{\text{valid}})$ is a discrete lattice in $\mathbb{R}^{12}$. Its differential geometry is analyzed through the smooth relaxation $\widetilde{\mathcal{M}}_{\text{flow}}$ (Section 2.2), defined as the zero set of the conservation residual Ψ below, subject to $x_2 \geq 1$, $0 < x_6 \leq 1.1 \times 10^8$, $x_{11} \geq x_2(20 \cdot \mathbb{I}_{x_1=6} + 8 \cdot \mathbb{I}_{x_1=17})$ and

$0 \le x_4 \le 1500 x_2$.

$$\Psi(x) = \begin{bmatrix} x_8 - \dfrac{10^6(x_4 + x_5)}{x_6} \\ x_9 - \dfrac{10^6(x_2 + x_3)}{x_6} \\ x_{10} - \dfrac{x_4}{x_2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \tag{8}$$

By the Regular Value Theorem, at any point where the Jacobian $J_\Psi(x) \in \mathbb{R}^{3\times 12}$ has full rank 3, the tangent space is $T_x\widetilde{\mathcal{M}}_{\text{flow}} = \ker J_\Psi(x)$. A physically realizable velocity $\vec{v} = [\dot{x}_0, \ldots, \dot{x}_{11}]^T$ therefore satisfies the linearized conservation laws:

$$\dot{x}_8 = \frac{10^6(\dot{x}_4 + \dot{x}_5)}{x_6} - \frac{x_8}{x_6}\dot{x}_6 \tag{9}$$

$$\dot{x}_9 = \frac{10^6(\dot{x}_2 + \dot{x}_3)}{x_6} - \frac{x_9}{x_6}\dot{x}_6 \tag{10}$$

$$\dot{x}_{10} = \frac{\dot{x}_4 - x_{10}\dot{x}_2}{x_2} \tag{11}$$

An adversary controls four elementary degrees of freedom, each generating a 1-parameter flow whose push-forward is a tangent vector field. Here $\vec{e}_k$ is the kth standard basis vector of $\mathbb{R}^{12}$ (indexed from 0):

$$\begin{aligned} \vec{v}_{\text{time}} &= \vec{e}_6 - \frac{x_8}{x_6}\vec{e}_8 - \frac{x_9}{x_6}\vec{e}_9, \\ \vec{v}_{\text{pad}} &= \vec{e}_4 + \frac{10^6}{x_6}\vec{e}_8 + \frac{1}{x_2}\vec{e}_{10}, \\ \vec{v}_{\text{chaff}} &= \vec{e}_2 + L_0\vec{e}_4 + \frac{L_0 \cdot 10^6}{x_6}\vec{e}_8 + \frac{10^6}{x_6}\vec{e}_9 + \frac{L_0 - x_{10}}{x_2}\vec{e}_{10} + H_0\vec{e}_{11}, \\ \vec{v}_{\text{win}} &= \mathbb{I}_{x_1=6}\ \text{sat}\left(\frac{W_{\text{target}} - x_7}{\epsilon_{\text{win}}}\right)\vec{e}_7, \end{aligned} \tag{12}$$

where temporal dilation adds delay without bytes or packets, payload padding adds forward dummy bytes, chaff injection adds forward zero-payload probes,

and window morphing moves the client window toward W_{target}. Here $\text{sat}(z) = \max(-1, \min(1, z))$ and $\epsilon_{\text{win}} > 0$. The probe sizes are $L_0 = 40$, $H_0 = 20$ for empty TCP segments per RFC 9293 §3.8.4 [37], $L_0 = 52$, $H_0 = 32$ with TCP timestamps [41], and $L_0 = 28$, $H_0 = 8$ for UDP [38]. Probes are classified at the ingress perimeter before server ACKs alter backward counts; bidirectional sessions are discussed in Section 6.1. The window field is globally Lipschitz rather than C^∞ and depends only on x_1 and x_7. The implementation (Supplementary Algorithm S1) uses its $\epsilon_{\text{win}} \to 0$ limit, $\mathbb{I}_{x_1=6}\,\text{sign}(W_{\text{target}} - x_7)\,\vec{e}_7$, which moves by $|W_{\text{target}} - x_7|$ and lands exactly on W_{target} without overshoot.

3.2. Commutativity, Invariants and the Lie Cone

The Lie bracket of vector fields is $[X, Y] = J_Y X - J_X Y$, where J_X and J_Y are their Jacobians.

Proposition 1 (Commutativity and Integrability of Physical Flow Generators)**.** *Because protocol $x_1 \in \{6, 17\}$ is an immutable attribute ($\dot{x}_1 \equiv 0$), the ambient manifold $\widetilde{\mathcal{M}}_{flow}$ decomposes into two disjoint smooth invariant sub-manifolds: $\widetilde{\mathcal{M}}^{(6)}_{flow}$ (TCP flows) and $\widetilde{\mathcal{M}}^{(17)}_{flow}$ (UDP flows). On each component, the indicator $\mathbb{I}_{x_1=6}$ is constant, and L_0, H_0 are constant along any trajectory (the implementation selects $H_0 = 32$ iff the mean forward header $x_{11}/x_2 \geq 26$; injecting chaff of header size H_0 moves that mean toward H_0 and never across the threshold, so the selection is invariant under the flow), and denominators satisfy $x_2 \geq 1, x_6 > 0$, ensuring $\vec{v}_{time}, \vec{v}_{pad}, \vec{v}_{chaff}$ are C^∞ vector fields and $\vec{v}_{win}$ is globally Lipschitz (its brackets exist almost everywhere and vanish there, because x_7 is neither read nor written by the other three generators). The Lie brackets between all four infinitesimal packet transformation*

generators vanish identically on $\widetilde{\mathcal{M}}_{flow}$*:*

$$[\vec{v}_i, \vec{v}_j] \equiv \mathbf{0} \in \mathbb{R}^{12}, \quad \forall i, j \in \{1, 2, 3, 4\} \tag{13}$$

Consequently, the generators span an Abelian Lie algebra $\mathfrak{g}_{flow} \cong \mathbb{R}^4$ *(or* $\mathbb{R}^3$ *for UDP).*

The proof of Proposition 1, and the Frobenius integrability corollary that follows from it (Supplementary Corollary S1), are given in the Supplementary Material (Section S1).

The admissible deformation cone is the positive span of the generators:

$$\mathcal{K}_{\text{net}}(x) = \left\{ \sum_{k=1}^{4} u_k \vec{v}_k(x) \;\middle|\; u_k \geq 0 \right\} \subset T_x \widetilde{\mathcal{M}}_{\text{flow}} \tag{14}$$

Theorem 1 (Universal Invariant Preservation)**.** *Let* $x_0 \in \widetilde{\mathcal{M}}_{flow}$ *be an initial network flow vector. For any measurable and locally bounded control signal* $u(t) \in L^{\infty}_{loc}(\mathbb{R}_{\geq 0}, \mathbb{R}^4_{\geq 0})$ *satisfying the admissible boundary rules* ($u_2(t) \leq (1500 - L_0)u_3(t)$ *whenever* $x_4(t) = 1500x_2(t)$*, and* $u_1(t) = 0$ *whenever* $x_6(t) = 1.1 \times 10^8$*), the trajectory* $x(t)$ *defined by the dynamical system:*

$$\dot{x}(t) = \sum_{k=1}^{4} u_k(t) \vec{v}_k(x(t)), \quad x(0) = x_0 \tag{15}$$

satisfies $x(t) \in \widetilde{\mathcal{M}}_{flow}$ *for all* $t \geq 0$*, with solutions existing uniquely and globally on* $[0, \infty)$*. Under discrete stepping, the continuous Lie cone guides directional descent while the closed-form analytical projection operator* $\mathcal{P}_{\mathcal{M}}$ *(Section 3.4) eliminates discrete* $\mathcal{O}(\Delta^2)$ *truncation drift and enforces packet and byte integrality* ($x_2 \in \mathbb{N}_{\geq 1}, x_4, x_6 \in \mathbb{N}$) *and MTU clamping* $x_4 \leq 1500x_2$*, ensuring that the discrete state sequence remains strictly in the physical packet manifold* $\mathcal{M}_{flow}$*.*

Proof sketch. Differentiating Ψ along the flow gives a linear homogeneous system $\dot{\Psi} = A(t)\Psi$ with diagonal, locally integrable $A(t)$, so $\Psi \equiv \mathbf{0}$ by Carathéodory's theorem [42]. The bounds $g_i \geq 0$ are affine, so the Nagumo–Brezis condition [43, 44] holds on every face and the manifold is forward invariant. The same theorem gives existence and uniqueness of global solutions. The full proof is in the Supplementary Material (Section S1).

3.3. Operational Cost and Optimal Control

The unit costs are $c_1 = 10^{-4}$ per second of induced delay, $c_2 = 10^{-2}$ per byte of padding, $c_3 = 1.0$ per probe packet, and $c_4 = 10^{-3}$ per normalized window displacement $s_4 = |W_{\text{target}} - x_7|/65535 \in [0, 1]$. The evasion problem is an optimal control problem:

$$\min_{u(t)\geq 0} \quad \int_0^{T^*} \left(\sum_{k=1}^{4} c_k u_k(t) \right) dt \tag{16}$$

$$\text{subject to} \quad \dot{x}(t) = \sum_{k=1}^{4} u_k(t) \vec{v}_k(x(t)), \quad x(0) = x_{\text{orig}} \tag{17}$$

$$F(x(T^*)) < 0 \quad \text{(Classifier Misclassification)} \tag{18}$$

Because the generators commute (Proposition 1), the displacement depends only on $s_k = \int_0^{T^*} u_k(t)\, dt \geq 0$, and the cost $\mathcal{C}(s) = \sum_{k=1}^{4} c_k s_k$ is linear and path-independent. The tree ensemble boundary $F(x) = 0$ is piecewise constant and non-convex, so FFE greedily follows the generator with the largest directional score decrease per unit cost, $-\nabla_k / c_k$, and finishes with boundary snapping (TSBS).

Algorithm 1 Flow-Field Evasion (FFE) Master Optimization Loop

Require: Classifier F, initial flow $x_0 \in \mathcal{M}_{\text{flow}}$, schema $\mathcal{S}$, step budget M, costs $\vec{c} = [c_1, \ldots, c_4]^T$, threshold $\tau = 0$
Ensure: Evasive flow $x^* \in \mathcal{M}_{\text{flow}}$ with $F(x^*) < \tau$, or FAILURE

$x \leftarrow \mathcal{P}_{\mathcal{M}}(x_0)$; $f \leftarrow F(x)$; $q \leftarrow 1$
for step $= 1$ **to** M **do**
  if $f < \tau$ **then**
    return $(x, \text{SUCCESS})$
  end if
  $\mathcal{D}, q_{\text{used}} \leftarrow \text{EstimateLieDerivatives}(F, x, \mathcal{S})$; $q \leftarrow q + q_{\text{used}}$ {Supplementary Algorithm S1}
  $k^* \leftarrow$ first maximizer of $(-\nabla_k)/c_k$ over $(k, \nabla_k, \vec{v}_k, \Delta s_k, f_k) \in \mathcal{D}$ with $\nabla_k < 0$; null if none
  if $k^* \neq$ null **then**
    $x \leftarrow \mathcal{P}_{\mathcal{M}}(x + \Delta s_{k^*}\, \vec{v}_{k^*})$; $f \leftarrow f_{k^*}$ {Cached probe, no extra query}
  else
    $x_{\text{snap}} \leftarrow \text{TSBS}(F, x, \mathcal{S})$
    if $F(x_{\text{snap}}) < f$ **then**
      $x \leftarrow x_{\text{snap}}$; $f \leftarrow F(x_{\text{snap}})$
    else
      $\Delta t \leftarrow \min(1.1 \times 10^8, \max(5 \times 10^4, 1.5\, x_6))$ {Fallback dilation; projection clamps duration at the 1.1×10^8 μs timeout}
      $x \leftarrow \mathcal{P}_{\mathcal{M}}(x + \Delta t\, \vec{v}_{\text{time}}(x))$; $f \leftarrow F(x)$; $q \leftarrow q + 1$
    end if
  end if
end for
if $f < \tau$ **then**
  return $(x, \text{SUCCESS})$
else
  return $(x, \text{FAILURE})$
end if

3.4. Algorithmic Realization of Flow-Field Evasion

Algorithm 1 gives the master loop of Flow-Field Evasion (FFE). Starting from a flagged flow $x_0 \in \mathcal{M}_{\text{flow}}$ with $F(x_0) \geq 0$, each iteration probes the four generators, takes the admissible step with the largest score reduction per unit cost, and falls back to boundary snapping or temporal dilation when no generator lowers the score.

Supplementary Algorithm S1 estimates the directional derivatives by prob-

ing the ensemble once per generator, since ∇F vanishes almost everywhere. The probe steps are $\Delta s_1 = \max(1000,\, 0.5x_6)$ μs, $\Delta s_2 = \max(50,\, 0.25x_4)$ bytes, $\Delta s_3 = \max(1,\, \mathrm{round}(0.5x_2))$ packets and $\Delta s_4 = |64240 - x_7|$, and the derivatives are taken per second of delay, per byte, per packet and per window unit of $\Delta s_4/65{,}535$, respectively. Every displacement is followed by the closed-form projection $\mathcal{P}_{\mathcal{M}}$ of Supplementary Algorithm S2, which restores immutable features, enforces monotonicity and integer counts, bounds header and packet lengths, and recomputes the coupled rates.

3.4.1. Tree-Split Boundary Snapping (TSBS)

Algorithm 2 implements Tree-Split Boundary Snapping (TSBS), whose candidates are generated by Supplementary Algorithm S3. Because the score is piecewise constant in each split feature, a small displacement cannot cross a split threshold. In Tier 1 white-box settings, TSBS therefore extracts the active split thresholds of the ensemble, keeps per feature the threshold closest to the benign side of the current value, and retains the projected candidate that most decreases F.

GBDTs also split on coupled telemetry features, so TSBS inverts their constraints in closed form. To cross an active byte-rate threshold θ_8 or packet-rate threshold θ_9, it lengthens the flow duration to

$$x_6^* = \frac{10^6(x_4 + x_5)}{\theta_8} + \epsilon_6 \quad \text{or} \quad x_6^* = \frac{10^6(x_2 + x_3)}{\theta_9} + \epsilon_6, \tag{19}$$

with $\epsilon_6 = 10\,\mu$s; the candidate is discarded if x_6^* exceeds the $1.1 \times 10^8\,\mu$s cap. To cross a forward mean packet length threshold θ_{10} from above, TSBS inverts $x_{10} = (x_4 + NL_0)/(x_2 + N)$ and adds the smallest integer chaff count

Algorithm 2 Tree-Split Boundary Snapping (TSBS)

Require: Classifier F, flow candidate $x \in \mathcal{M}_{\text{flow}}$ with $F(x) \geq \tau$ (still flagged malicious), feature schema $\mathcal{S}$, tolerance ϵ_{rel}
Ensure: Snapped flow $x^* \in \mathcal{M}_{\text{flow}}$

$x^* \leftarrow \text{copy}(x)$
active_splits $\leftarrow$ ExtractActiveSplits(F, x^*)
$\mathcal{C} \leftarrow$ GenerateCandidates$(x^*, \text{active_splits}, \epsilon_{\text{rel}})$ {Boundary-crossing pool (Supplementary Algorithm S3)}
{Evaluate projected candidates and retain best partition crossing}
$x_{\text{best}} \leftarrow x^*$; $s_{\text{best}} \leftarrow F(x^*)$
for each $x_c \in \mathcal{C}$ **do**
  $x_{\text{proj}} \leftarrow$ ProjectManifoldInvariants$(x_c, \mathcal{S})$
  if $F(x_{\text{proj}}) < s_{\text{best}}$ **then**
    $s_{\text{best}} \leftarrow F(x_{\text{proj}})$; $x_{\text{best}} \leftarrow x_{\text{proj}}$
    if $s_{\text{best}} < 0$ **then**
      break {Early exit once benign classification is achieved}
    end if
  end if
end for
return x_{best}

that satisfies the inequality,

$$N_{\text{chaff}} = \left\lfloor \frac{x_4 - \theta_{10} x_2}{\theta_{10} - L_0} \right\rfloor + 1 \qquad (L_0 < \theta_{10} < x_{10}). \tag{20}$$

For TCP flows, TSBS also snaps the window advertisement toward the benign OS baseline $x_7^* = 64{,}240$ and toward adjacent split partitions.

3.4.2. Query Accounting

Executing all $M_{\max} = 20$ iterations without early termination costs at most $Q_{\max} \leq 4M_{\max} + 1 = 81$ model queries for the baseline query and Lie-derivative probes, or ≤ 101 if stagnation fallbacks fire, plus the TSBS candidate evaluations, which are not covered by this bound; UDP flows and TCP flows whose window already equals the benign value skip window morphing and cost $q = 3$ probes per iteration. In practice, query complexity depends directly on the physical flow profile of each intrusion family: localized

attacks such as low-and-slow infiltration and web application exploits evade with high efficiency (6.5 and 12.4 queries per flow on average), while complex volumetric flows navigating tighter decision corridors require additional probes, yielding an overall mean of 79.3 queries on the master benchmark cohort (and 84.4 queries in ablation sweeps). This figure is deliberately conservative: a Tier 1 adversary could evaluate candidate TSBS split crossings offline on its local copy of the ensemble, but our evaluation counts every probe and boundary candidate evaluation as an active model query, as a strictly black-box adversary would have to. Because $K \leq 4$ and $M \leq 20$ are constants, each iteration costs $\mathcal{O}(T \cdot D)$ tree evaluations, and Table 1 compares this asymptotic cost with exact MILP and heuristic search.

4. Experiments and Results

4.1. Experimental Setup and Benchmark Corpora

Evaluations on a single testbed have limited external validity [7, 23, 24, 25], and detectors trained on one environment transfer poorly to another [45, 46, 47]. We therefore evaluate on real flow traffic extracted from three widely studied benchmark datasets: CSE-CIC-IDS2018 (enterprise AWS infrastructure) [1], UNSW-NB15 (contemporary hybrid network environment) [14], and CIC-IDS2017 (Canadian Institute for Cybersecurity) [1]. Each benchmark corpus consists of 12,000 real network flows extracted directly from official repositories (the AWS S3 repository for CSE-CIC-IDS2018 and official public archives for UNSW-NB15 and CIC-IDS2017), mapped onto our 12-feature physical flow schema.

Each corpus contains 12,000 flows, of which 9,000 are benign and 3,000

malicious. The malicious flows form five families of 1,000, 500, 700, 500 and 300 flows: a SYN flood, a UDP flood, brute-force authentication against SSH and FTP, SQL injection and cross-site scripting, and low-and-slow command-and-control infiltration. The data are split into 8,000 training, 1,000 validation and 3,000 test flows. The threat model is an ingress perimeter that inspects forward telemetry before bidirectional session termination (Section 2.2).

The targets are XGBoost (CUDA histogram method) and LightGBM, each a 60-tree gradient-boosted ensemble of depth 6 with learning rate 0.1. Experiments ran on an NVIDIA RTX 5000 Ada GPU with 16 GB of memory, using Python 3.12, XGBoost 3.4.1, LightGBM 4.7.0, CatBoost 1.2.10 and scikit-learn 1.8.0.

We compare five evasion methods against FFE:

- Naive coordinate descent in $\mathbb{R}^d$ with a fixed step $\Delta = 0.05\,\sigma_f$ and no immutability, monotonicity or rate-coupling checks.
- ZOO-NIDS [9]: zeroth-order coordinate optimization with randomly sampled coordinates, symmetric difference quotients, Adam updates and box bounds.
- HopSkipJump-NIDS [10]: a decision-based boundary attack with geometric boundary projections.
- CAA/MOEVA [12, 13]: multi-objective evolutionary search (population 25, 20 generations) optimizing detection probability and perturbation cost over unconstrained feature space.

- Traffic Manipulator PSO [11]: particle swarm optimization over packet intervals and forward dummy bytes (15 particles, 25 iterations, up to 375 evaluations per flow).
- FFE: the Lie-steering procedure (Algorithms 1–2) with step budget $M = 20$.

ASR is the fraction of attacks with $P(\text{Malicious} \mid x^*) < 0.5$. PFR is the fraction of candidate flows that pass the protocol sanity checks, the non-negativity checks and the exact coupled rate equations, that is, $x^* \in \mathcal{M}_{\text{flow}}$. For unconstrained baselines that never produce an evasive candidate, PFR is computed over evasive candidates only. Real ASR counts an attack only when it both evades and is physically feasible:

$$\text{Real ASR} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}\left[F(x_i^*) < 0 \ \wedge \ x_i^* \in \mathcal{M}_{\text{flow}}\right], \tag{21}$$

where F is the detector score and negative values denote evasion. Average queries counts the model evaluations per sample, and total time is the wall-clock time for the cohort. The perturbation cost is the mean feature-normalized L_1 distance over the $d = 12$ features:

$$\bar{D}_{L_1}(x^*, x) = \frac{1}{d} \sum_{j=1}^{d} \frac{|x_j^* - x_j|}{\sigma_j}, \tag{22}$$

where σ_j is the empirical standard deviation of feature j in the training corpus.

To make exhaustive iterative optimization computationally tractable across all baselines, architectures, and multi-step sweeps, each benchmark corpus uses a stratified sample of 12,000 flows (9,000 benign, 3,000 attack across

five families) extracted from the multi-gigabyte raw repositories. Each attack method is evaluated on the same master cohort of 100 correctly classified malicious flows sampled at random from the test partition. The per-family and cross-dataset cohorts also evaluate 100 flows each. The ablation and defense hardening studies operate on the cohort of detected malicious test flows ($N = 95$ detected flows across all five families, $\sim$20 flows per family), accounting for slight variations from the master cohort (e.g., 84.4 vs. 79.3 queries at $M = 20$). Table 5 also trains CatBoost and Random Forest models on the same corpus. Each study trains its own model instances, so clean accuracy for the same architecture differs slightly across tables (98.60% in Table 3, 98.62% in Table 6). All experiments use a fixed random seed for deterministic reproducibility across baseline runs; multi-seed variance and full un-sampled trace evaluation are discussed as threats to validity in Section 6.1.

4.2. Master Comparative Benchmark

Table 2: Adversarial evasion benchmark against CUDA XGBoost on real CSE-CIC-IDS2018 flow traffic ($N = 100$ attacks).

Method	ASR (%)	PFR (%)	Real ASR (%)	Avg Queries	Avg Cost ($\bar{D}_{L_1}$)	Total Time (s)
Unconstrained Feature-Space	39.0	5.0	4.0	77.8	9,673.78	6.62
ZOO-NIDS [9]	8.0	6.0	0.0	144.0	9,323.43	30.04
HopSkipJump-NIDS [10]	100.0	0.0	0.0	47.0	1.64×10^6	10.55
CAA / MOEVA [12, 13]	39.0	15.0	1.0	336.2	111.85	103.31
Traffic Manipulator PSO [11]	54.0	100.0	54.0	182.6	2,946.18	39.85
FFE (ours)	**59.0**	**100.0**	**59.0**	**79.3**	**0.392**	**28.66**

Costs and metric definitions are given in Section 4.

Table 2 compares FFE with the five baselines on real network flows. Several baselines evade the detector mathematically but produce physically unrealizable flow telemetry. For example, HopSkipJump-NIDS achieves 100.0%

unconstrained ASR, yet achieves 0.0% Physical Feasibility Rate (PFR) and 0.0% Real ASR because it projects candidate flows into non-physical coordinates with an astronomical perturbation cost ($\bar{D}_{L_1} = 1.64 \times 10^6$). Similarly, ZOO-NIDS and CAA/MOEVA achieve only 0.0% and 1.0% Real ASR (with PFRs of 6.0% and 15.0%), demonstrating that unconstrained mutations severely decouple conservation laws and transport header invariants.

Both Traffic Manipulator PSO and Flow-Field Evasion operate strictly within valid physical flow semantics, achieving 100.0% PFR and reaching comparable evasion rates on CSE-CIC-IDS2018 (59.0% for FFE vs. 54.0% for PSO). However, their operational footprints diverge by orders of magnitude. Because PSO optimizes blind particle velocities without knowledge of tree decision geometries, it requires 2.3× higher query complexity (182.6 vs. 79.3 queries) and incurs an astronomical $7,500\times$ higher perturbation overhead ($\bar{D}_{L_1} = 2,946.18$ vs. 0.392), aggressively injecting chaff packets and delays. In contrast, FFE navigates along infinitesimal Lie generators and snaps to the nearest boundary partition via TSBS, achieving extreme perturbation parsimony and query efficiency. FFE's primary advantage over packet-space heuristics therefore lies in surgical geometric boundary traversal rather than brute-force evasion margins.

4.3. Per-family and cross-dataset validation

Table 3 reports FFE by attack family on CSE-CIC-IDS2018 (Panel A) and by dataset and target model (Panel B).

As shown in Panel A, FFE exhibits distinct physical evasion behavior depending on the flow geometry of each attack family. Low-and-slow infiltration (100.0% Real ASR, 6.5 queries) and web application exploits (100.0% Real

Table 3: Flow-Field Evasion across attack families on CSE-CIC-IDS2018 against CUDA XGBoost (Panel A, $N = 100$ flows per family) and across datasets and target models (Panel B, $N = 100$ flows per row).

Panel A: attack family breakdown (CSE-CIC-IDS2018, XGBoost)

Attack family	Real ASR (%)	PFR (%)	Avg Queries	Avg Latency (ms)
Family 1: Volumetric DoS SYN Flood	0.0	100.0	175.0	897.4
Family 2: UDP Volumetric Flood	96.0	96.0	30.3	183.6
Family 3: Brute Force Authentication	0.0	100.0	175.0	915.3
Family 4: Web Application Exploits	100.0	100.0	12.4	74.2
Family 5: Low-and-Slow Infiltration	100.0	100.0	6.5	40.3
Unweighted mean (5 families)	**59.2**	**99.2**	**79.8**	**422.2**

Panel B: cross-dataset and model architecture validation

Dataset	Model	Clean Acc (%)	Clean AUC (%)	Real ASR (%)	PFR (%)	Queries	Latency (ms)
CSE-CIC-IDS2018	XGBoost	98.60	99.87	59.0	100.0	79.3	417.0
CSE-CIC-IDS2018	LightGBM	98.67	99.85	97.0	98.0	10.4	15.5
UNSW-NB15	XGBoost	94.60	98.76	98.0	100.0	32.2	194.9
UNSW-NB15	LightGBM	94.77	98.70	94.0	100.0	34.3	61.5
CIC-IDS2017	XGBoost	99.87	100.00	11.0	93.0	170.3	986.1
CIC-IDS2017	LightGBM	99.90	100.00	3.0	93.0	213.7	381.3

ASR, 12.4 queries) evade with high efficiency via window morphing and chaff injection. UDP volumetric floods achieve 96.0% Real ASR at 30.3 queries via rate dilution ($\vec{v}_{\text{time}}$). Conversely, DoS SYN floods and SSH/FTP brute force present strict physical hurdles on CSE-CIC-IDS2018 due to deep tree splits on destination port (which is immutable) and fixed packet length bounds, requiring additional budget.

Panel B demonstrates the cross-dataset validity of FFE across three independent benchmark datasets and two leading gradient boosting implementations (XGBoost and LightGBM). On UNSW-NB15, FFE achieves 98.0% and 94.0% Real ASR on XGBoost and LightGBM, respectively, with moderate query budgets (32.2 and 34.3 queries). On CSE-CIC-IDS2018, LightGBM is substantially more vulnerable than XGBoost (97.0% vs. 59.0% Real ASR), requiring only 10.4 queries per flow and 15.5 ms latency. On CIC-IDS2017, where the tree ensembles achieve near-perfect discrimination (99.87%–99.90% clean accuracy and 100.00% AUC), Real ASR drops to 11.0% (XGBoost) and

3.0% (LightGBM). This lower evasion rate stems directly from the feature topology of the CIC-IDS2017 capture: benign and attack flows are cleanly separated by immutable destination ports and immutable backward packet characteristics, leaving narrow continuous corridors for forward Lie steering. This transparent cross-dataset comparison demonstrates that tree ensemble vulnerability to problem-space evasion is fundamentally bounded by feature distributions and split depths rather than inherent classifier invulnerability. Across all datasets and models, CUDA XGBoost incurs higher per-sample latency than LightGBM due to host-to-device memory transfers on single-row inference.

4.4. Ablations and transferability

Table 4 reports three ablation sweeps on the CUDA XGBoost target on real CSE-CIC-IDS2018 flow traffic, using a balanced cohort of $N = 95$ detected malicious flows across all five families. Panel A sweeps the step budget M, Panel B isolates generator subsets, and Panel C compares operational cost regimes.

In Panel A, ASR rises monotonically with step budget, from 14.7% at $M = 1$ (5.0 queries, $\bar{D}_{L_1} = 0.165$) to 51.6% at $M = 5$ (28.1 queries), reaching a saturation plateau of 56.8% at $M = 10$ (49.8 queries, $\bar{D}_{L_1} = 0.375$). Increasing the budget beyond $M = 10$ yields no further evasion on this cohort because the remaining unevaded flows are bounded by immutable port and protocol constraints. Physical feasibility remains 98.9% across all budgets.

Panel B demonstrates the necessity of the complete Lie cone. Individual generators achieve limited evasion: temporal dilation alone reaches 25.3% (adding 77.3 s of flow duration), chaff injection reaches 25.3%, window mor-

phing reaches 16.8%, and payload padding reaches 12.6%. Combining generators produces synergistic gains: Time + Chaff achieves 45.3%, while the Complete Lie Cone achieves 56.8% ASR at 84.4 queries. Notably, Padding + Chaff achieves 36.8% ASR with zero added flow delay (0.00 s) at only 10.4 queries per flow.

Panel C examines the effect of control cost weighting $\vec{c}$. Across all four operational regimes, FFE consistently achieves 56.8% ASR with comparable query counts (83.9–84.4 queries), with $\vec{v}_{\text{time}}$ serving as the dominant generator in 85%–86% of gradient steps. Sweeping the TSBS relative snapping offset ϵ_{rel} from 0 to 1.0 similarly maintains 56.8% ASR and 98.9% PFR, confirming stability across threshold snapping tolerances.

In many operational settings the attacker cannot query the deployed model directly. Table 5 therefore evaluates black-box cross-architecture transferability by measuring the attack success rate of flows crafted on a source tree ensemble and evaluated against three independent target architectures ($N = 95$ detected flows).

The empirical transferability matrix reveals strong structural cross-vulnerability among tree ensembles on real network traffic. Flows crafted against XGBoost (CUDA) transfer with exceptional effectiveness to LightGBM (95.8% ASR) and Random Forest (77.9% ASR), matching CatBoost at 56.8%. Conversely, LightGBM is universally the most susceptible target architecture to black-box transfer, suffering 80.0%–95.8% evasion when attacked by adversarial flows crafted on other ensembles. Flows crafted on CatBoost transfer robustly to both LightGBM (87.4%) and Random Forest (76.8%). Transfers into XGBoost range from 25.3% to 42.1%, demonstrating that XGBoost's

Table 4: Ablation studies of Flow-Field Evasion on CUDA XGBoost using real CSE-CIC-IDS2018 flow traffic ($N = 95$ detected flows, balanced across families; PFR = 98.9% across all rows).

Panel A: step budget sweep			
Step budget M	**ASR (%)**	**Avg queries**	**Avg cost $\bar{D}_{L_1}$**
$M = 1$	14.7%	5.0	0.165
$M = 2$	31.6%	10.7	0.285
$M = 3$	35.8%	16.5	0.290
$M = 5$	51.6%	28.1	0.301
$M = 10$	56.8%	49.8	0.375
$M = 20$	56.8%	84.4	0.408
Panel B: generator subset ablations ($M = 20$)			
Generator subset	**ASR (%)**	**Avg queries**	**Added duration (s)**
Time only ($\vec{v}_{\text{time}}$)	25.3%	57.4	77.34
Padding only ($\vec{v}_{\text{pad}}$)	12.6%	4.1	0.00
Chaff only ($\vec{v}_{\text{chaff}}$)	25.3%	6.4	0.00
Window only ($\vec{v}_{\text{win}}$)	16.8%	2.8	0.00
Time + padding	37.9%	73.2	63.22
Time + chaff	45.3%	80.5	63.21
Padding + chaff	36.8%	10.4	0.00
Complete Lie cone	**56.8%**	**84.4**	**50.04**
Panel C: operational cost regimes ($M = 20$)			
Cost regime	**ASR (%)**	**Avg queries**	**Mean delay (s)**
A (default / balanced): $\vec{v}_{\text{time}}$, 86%	56.8%	84.4	50.04
B (latency-sensitive): $\vec{v}_{\text{time}}$, 86%	56.8%	84.2	49.96
C (unconstrained delay): $\vec{v}_{\text{time}}$, 85%	56.8%	84.1	50.04
D (uniform / flat): $\vec{v}_{\text{time}}$, 85%	56.8%	83.9	49.97

split thresholds on discrete packet length bounds establish narrower transfer corridors than histogram-based leaf-wise trees.

5. Defensive Countermeasures

Standard boosting chooses split features by training-loss reduction alone, so features that separate the classes widely in benchmark corpora, such as duration and the rates, sit near the root although an attacker can shift them

Table 5: Cross-architecture adversarial transferability matrix on real CSE-CIC-IDS2018 traffic (Attack Success Rate % across $N = 95$ detected flows).

Source \ Target	XGBoost (CUDA)	LightGBM	CatBoost	Random Forest
XGBoost (CUDA)	**56.8%**	95.8%	56.8%	77.9%
LightGBM	25.3%	**98.9%**	12.6%	30.5%
CatBoost	42.1%	87.4%	**95.8%**	76.8%
Random Forest	35.8%	80.0%	14.7%	**91.6%**

by delaying or padding packets. We evaluate malleability-penalized splitting (MPS), which discourages splits on malleable features, and Lie-guided adversarial training (LGAT), which trains on flows generated along the Lie generators.

5.1. Malleability-penalized tree splitting

Following the robust split formulations of Chen et al. [32] and Calzavara et al. [34], the split score for feature j is regularized by its physical malleability index.

Definition 1 (Feature Physical Malleability Index)**.** The physical malleability index $M_{\text{mal}}(j)$ of feature j is defined as the empirical expectation over the training distribution of the maximum dimensionless variation induced by the Lie Cone $\mathcal{K}_{\text{net}}$:

$$M_{\text{mal}}(j) = \mathbb{E}_{x \sim \mathcal{D}_{\text{train}}} \left[\max_{k \in \{1,\dots,4\}} \frac{\delta s_k |\vec{v}_k(x)_j|}{\sigma_j} \right] \tag{23}$$

where σ_j is the empirical standard deviation of feature j across the training corpus $\mathcal{D}_{\text{train}}$, and δs_k denotes the characteristic single-step perturbation scale for generator k ($\delta s_1 = 10^4\,\mu\text{s}, \delta s_2 = 100\,\text{B}, \delta s_3 = 1\,\text{pkt}, \delta s_4 = 10^3\,\text{B}$).

The robust split selection objective is

$$\text{Gain}_{\text{robust}}(j, \theta) = \Delta\mathcal{L}(j, \theta) - \alpha_{\text{mal}} \cdot \overline{\Delta\mathcal{L}} \cdot M_{\text{mal}}(j) \tag{24}$$

where $\Delta\mathcal{L}(j,\theta)$ is the impurity gain of split (j,θ), $\overline{\Delta\mathcal{L}}$ is the baseline mean impurity gain across candidate splits at the node, ensuring dimensional consistency, and $\alpha_{\text{mal}} \geq 0$ is a dimensionless hyperparameter governing the trade-off between clean training loss and malleability robustness.

The experiments do not optimize this objective. The MPS model instead masks duration (x_6) and the two rate features (x_8, x_9) to zero in the training, test and attack inputs, which is the $\alpha_{\text{mal}} \to \infty$ limit for those features. The mask is fixed by hand rather than computed from M_{mal}, and other features are unpenalized. Because the masked columns are constant, trees cannot split on them, so the attacker affects MPS models only through unmasked features.

5.2. Lie-guided adversarial training

Standard adversarial training assumes an L_p ball around each input [48], which in network security generates physically invalid samples that degrade clean accuracy [35, 7]. LGAT instead restricts the inner maximization to the physical flow manifold, solving

$$\min_{\Theta} \frac{1}{N} \sum_{i=1}^{N} \max_{u \in \mathcal{U}_{\text{budget}}} \mathcal{L}\left(F_{\Theta}\left(\mathcal{A}_{\text{cone}}\left(x_i, u\right)\right), y_i\right) \tag{25}$$

where $\mathcal{A}_{\text{cone}}(x,u)$ is the flow reached from x along the integral curves of the Lie generators under the control u, and $\mathcal{U}_{\text{budget}}$ is the set of nonnegative controls whose cost $\sum_{k=1}^{4} c_k \int_0^{T_{\max}} u_k(\tau)\, d\tau$ does not exceed Γ.

Supplementary Algorithm S4 gives the general scheme, in which malicious flows are replaced by FFE outputs at step budget $M_{\text{train}} = 5$ in every boosting round, with $\beta = 0.5$. The experiments use a single-round instance: adversarial examples are generated once, against the undefended model, for

a random 50% of the malicious training flows and appended to the training set, and the ensemble uses standard split scoring.

5.3. Evaluation

We compared an undefended XGBoost model with three hardened variants on real CSE-CIC-IDS2018 flow traffic, each with 60 trees of depth 6 and learning rate 0.1. All models are evaluated on the cohort of $N = 95$ detected malicious flows, using FFE at step budget $M = 5$ (Table 6).

Table 6: Defensive hardening against Flow-Field Evasion on real CSE-CIC-IDS2018 flow traffic (XGBoost; cohort $N = 95$; attack evaluation budget $M = 5$; all values in %). Robust recall is 100 − ASR.

Model	Clean acc.	Clean AUC	Robust recall	FFE ASR
Undefended	98.62	99.88	48.4	51.6
MPS-hardened	98.67	99.85	1.1	98.9
LGAT-hardened	98.72	99.85	74.7	25.3
MPS + LGAT	98.67	99.84	94.7	5.3

Combining MPS with LGAT dramatically suppresses FFE evasion from 51.6% to 5.3% and raises robust recall from 48.4% to 94.7%, with virtually zero degradation in clean classification performance (98.67% accuracy and 99.84% ROC-AUC, compared to 98.62% and 99.88% undefended).

The Malleability Penalty Trap: Why Standalone MPS Backfires. A striking empirical phenomenon in Table 6 is that standalone MPS severely degrades model robustness: FFE evasion surges from 51.6% on the undefended model to 98.9% on the MPS-hardened model (robust recall collapses to 1.1%). This counter-intuitive vulnerability arises from what we term the *malleability penalty trap*. When split gain penalties heavily suppress malleable temporal

features (flow duration x_6, flow bytes/s x_8, and flow packets/s x_9), standard greedy tree induction must find alternative partitions to minimize empirical training loss. The ensemble is consequently forced to place its primary decision splits on unpenalized features, notably forward packet counts (x_2), forward payload bytes (x_4), transport header length (x_{11}), and TCP window size (x_7).

Crucially, because these alternative features are not regularized by adversarial examples, the resulting decision boundaries sit immediately adjacent to natural attack clusters. For an adversary equipped with FFE, this boundary migration drastically simplifies the evasion path: instead of having to expend substantial query budgets diluting coupled rate equations via temporal pacing ($\vec{v}_{\text{time}}$), the attacker can execute instantaneous boundary snaps (TSBS) using infinitesimal chaff packet injection ($\vec{v}_{\text{chaff}}$), payload padding ($\vec{v}_{\text{pad}}$), or TCP window adjustment ($\vec{v}_{\text{win}}$) with near-zero cost in only 1–3 queries. Standalone feature penalization therefore creates an acute blind spot by shifting splits onto unfortified mutable dimensions.

In contrast, Lie-guided adversarial training (LGAT) exposes the trees to valid Lie cone trajectories during induction, halving evasion to 25.3%. When integrated in the dual-hardened configuration, MPS successfully prevents the ensemble from learning spurious shortcuts on rate and timing telemetry, while LGAT simultaneously inoculates the remaining discrete feature splits against Lie cone boundary traversal. This synergy demonstrates that problem-space adversarial training cannot rely on heuristic feature masking alone, but requires manifold-aware data augmentation to achieve certified robustness without sacrificing clean detection fidelity.

6. Discussion and Conclusions

6.1. Operational Feasibility, Query Efficiency, and Limitations

Feature-space attacks are fast, but their output is not guaranteed to be a physically realizable flow (Section 2). FFE pays for feasibility with a different structure. Its generators act on the flow manifold, so trajectories in the continuous Lie cone satisfy the conservation and rate-coupling invariants exactly, and a projection step restores them on the discrete flow schema. Feasibility is therefore 100% in every experiment, but this figure measures conformance to the 12-feature schema, not an independent test on live packets.

Against population and swarm methods, the primary advantage is not an overwhelming evasion rate, but query and perturbation parsimony. Traffic Manipulator PSO achieves competitive evasion (54.0% vs. 59.0% Real ASR on CSE-CIC-IDS2018), yet it requires 182.6 queries and incurs an astronomical perturbation cost ($\bar{D}_{L_1} = 2,946.18$), blindly injecting massive chaff packets and temporal delays. In contrast, FFE achieves 59.0% at $\bar{D}_{L_1} = 0.392$ with 79.3 queries (and under 13 queries on targeted exploit cohorts). The saving comes from the ensemble's own geometry: TSBS snaps flows onto adjacent split boundaries rather than searching through stochastic dispersion. Compared with MILP and SAT/SMT encodings, FFE does not encode the whole ensemble as a constraint problem, whose worst-case cost is $2^{O(DT)}$ for MILP with T trees of depth D. It also accepts nonlinear invariants such as $x_8 = 10^6(x_4 + x_5)/x_6$, which a purely linear encoding cannot express.

The method has clear limits. TSBS requires Tier-1 white-box access to the split thresholds. Under Tier-2 access the attack depends on probes and transfer, where source models like XGBoost achieve 56.8%–95.8% transfer-

ability across target architectures (reaching 95.8% against LightGBM and 77.9% against Random Forest). Traffic Manipulator operates on captured PCAP files, whereas our packet realization is a Scapy/NFQueue proxy that has not been replayed at line rate. Control is restricted to forward traffic, because backward server features are treated as immutable. Deep packet inspection is out of scope, since payload padding alone does not defeat signature matches. Rate dilution also lowers the intensity of volumetric floods, so an attacker trades burst rate for stealth.

Evasion efficacy also exhibits significant dataset dependency: while FFE reaches 59.0%–98.0% Real ASR on CSE-CIC-IDS2018 and UNSW-NB15, evasion drops to 3.0%–11.0% on CIC-IDS2017. On CIC-IDS2017, trees partition classes along immutable destination ports and backward packet statistics, leaving narrow continuous corridors for forward Lie steering. This confirms that tree ensemble robustness is governed by underlying telemetry distributions and split depth rather than intrinsic classifier invulnerability.

Threats to validity include our evaluation on 12,000-flow stratified samples extracted from multi-gigabyte raw captures, the evaluation of defenses on the detected test cohort ($N = 95$ flows), and the use of fixed random seeds without variance intervals. While sampling and fixed seeds enable exhaustive per-flow baseline comparisons within computational limits, scaling to full unsampled wire traces and multi-seed variance audits remains essential future work.

Each generator has a protocol-compliant packet-level realization: sender pacing for temporal dilation, padding in protocol fields or ignored application messages such as `SSH_MSG_IGNORE` sent before encryption [39], payload-free

pure ACK or keep-alive chaff [37, 41] or zero-length UDP datagrams [38], and window adjustment during the handshake. A proxy then updates sequence numbers and recomputes the IPv4 and transport checksums [36, 37, 38].

6.2. Concluding Remarks and Future Directions

This article has demonstrated that tree-ensemble NIDS reporting near-perfect benchmark accuracy are vulnerable to evasion that respects flow physics. Flow-Field Evasion models four packet-level actions as infinitesimal generators on the flow manifold. Trajectories in the continuous Lie cone preserve conservation invariants, and a projection enforces them on the discrete schema. Evaluated on real flow traffic from CSE-CIC-IDS2018, UNSW-NB15, and CIC-IDS2017, FFE reaches 59.0% Real ASR against XGBoost at an ultra-low perturbation cost of 0.392 with 100% physical feasibility, matching Traffic Manipulator PSO in evasion efficacy while slashing perturbation cost by 7,500× and query complexity by more than half. Across architectures, FFE reaches up to 94.0%–98.0% Real ASR on UNSW-NB15 and LightGBM, but is constrained to 3.0%–11.0% on CIC-IDS2017 due to deep immutable feature splits. Crucially, standalone feature malleability penalization creates a vulnerability trap (driving evasion to 98.9% by shifting boundaries to unpenalized features), whereas dual hardening (MPS + LGAT) successfully suppresses FFE evasion to 5.3% (94.7% robust recall) with zero clean accuracy degradation. Future work will investigate line-rate replay on hardware testbeds and certified robustness bounds for adversarially trained ensembles.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Data availability

The code, dataset extraction scripts, and reproducible experimental pipelines, together with all benchmark result archives for FFE, the baseline methods (Naive, ZOO-NIDS, HopSkipJump-NIDS, CAA/MOEVA, and Traffic Manipulator PSO), ablations, transferability, and defenses, are publicly available at https://github.com/vinhqdang/adversarial_ids. All flow data originate from public benchmark archives (CSE-CIC-IDS2018, UNSW-NB15, and CIC-IDS2017).

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